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JAYAWANT SHIKSHAN PRASARAK MANDAL'S

Jayawantrao Sawant College of Engineering

T.E. Computer Engineering (2026-27)

Computer Network System (CCE Activity–1)

A Practical Network Implementation Report

CAMPUS NETWORK DESIGN, SIMULATION, AND TROUBLESHOOTING USING CISCO PACKET TRACER

Prepared By:

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1. Aim

To design, simulate, configure, test, and troubleshoot a campus network using Cisco Packet Tracer.

2. Objectives

- Design a small campus network using a router, switches, end devices, a server, and a printer.
- Connect devices using appropriate Ethernet connections.
- Configure two separate IP networks and their default gateways.
- Configure router interfaces and verify their operational status.
- Test connectivity using the ping command.
- Demonstrate inter-network communication through the router.
- Identify and troubleshoot an incorrect default gateway.
- Save and verify the final network configuration.

3. Software / Devices Required

Sr. No.	Device / Software	Quantity
1	Cisco Packet Tracer	1
2	Cisco 2911 Router	1
3	Cisco 2960 Switch	2
4	PC	4
5	Server	1
6	Printer	1
7	Copper Straight-Through Cable	8

4. Network Topology

The campus network is divided into two IP networks. The router provides communication between the two networks. Switch0 serve

```
Router
 /     \
G0/0   G0/1
 /       \
Switch0   Switch1
 / | \   / | \
PC0 PC1 Server   PC2 PC3 Printer
```

5. IP Addressing Scheme

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router	G0/0	192.168.10.1	255.255.255.0	—
Router	G0/1	192.168.20.1	255.255.255.0	—
PC0	FastEthernet0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	FastEthernet0	192.168.10.3	255.255.255.0	192.168.10.1
Server	FastEthernet0	192.168.10.10	255.255.255.0	192.168.10.1
PC2	FastEthernet0	192.168.20.2	255.255.255.0	192.168.20.1
PC3	FastEthernet0	192.168.20.3	255.255.255.0	192.168.20.1
Printer	FastEthernet0	192.168.20.10	255.255.255.0	Not available on selected printer model

6. Device Connections

Sr. No.	Device / Port	Connected To
1	Router G0/0	Switch0 Fa0/1
2	Router G0/1	Switch1 Fa0/1
3	PC0 Fa0	Switch0 Fa0/2
4	PC1 Fa0	Switch0 Fa0/3
5	Server Fa0	Switch0 Fa0/4
6	PC2 Fa0	Switch1 Fa0/2
7	PC3 Fa0	Switch1 Fa0/3
8	Printer Fa0	Switch1 Fa0/4

7. Step-by-Step Implementation

7.1 Create the topology

Added one Cisco 2911 router, two Cisco 2960 switches, four PCs, one server, and one printer.

7.2 Connect devices

Used Copper Straight-Through cables for the router-switch and end-device-switch connections.

7.3 Configure Router G0/0

Assigned 192.168.10.1/24 to G0/0 and enabled the interface.

7.4 Configure Router G0/1

Assigned 192.168.20.1/24 to G0/1 and enabled the interface.

7.5 Configure end devices

Assigned the IP addresses, subnet masks, and default gateways shown in the addressing table.

7.6 Verify interfaces

Used show ip interface brief to confirm both router interfaces were up/up.

7.7 Verify configuration / 8. Verification and Testing

7.7 Verify configuration

Used show running-config to inspect the saved interface configuration.

7.8 Test connectivity

Used ping to verify same-network and inter-network communication.

7.9 Router Configuration Commands

enable
configure terminal
interface gigabitEthernet0/0
ip address 192.168.10.1 255.255.255.0
no
shutdow
n exit
interface gigabitEthernet0/1
ip address 192.168.20.1 255.255.255.0
no
shutdow
n exit
en
d
writ
e

8. Verification and Testing

The following tests were performed during the implementation:

Test	Source → Destination	Observed Result	Status
PC0 → PC1	192.168.10.2 → 192.168.10.3	4 packets received, 0% loss	Successful
PC0 → Server	192.168.10.2 → 192.168.10.10	4 packets received, 0% loss	Successful
PC0 → PC2	192.168.10.2 → 192.168.20.2	3 replies after first ARP-related timeout	Successful
PC2 → Printer	192.168.20.2 → 192.168.20.10	4 packets received, 0% loss	Successful

8.1. Verification Command

show ip interface brief

Expected final state:

GigabitEthernet0/0
192.168.10.1 up up
GigabitEthernet0/1
192.168.20.1 up up

9. Troubleshooting Demonstration

A deliberate default-gateway error was introduced on PC2 to demonstrate troubleshooting.

Item	Details
Problem introduced	PC2 default gateway changed from 192.168.20.1 to 192.168.20.100.
Test performed	ping 192.168.10.2 from PC2.
Observed result	Request timed out.
Cause	PC2 had an incorrect default gateway, so it could not correctly forwar d traffic to the other network.
Solution	Changed the default gateway back to 192.168.20.1.
Verification	The same ping then returned replies, confirming that communication was restored.

10. Evidence / Screenshots (continued)

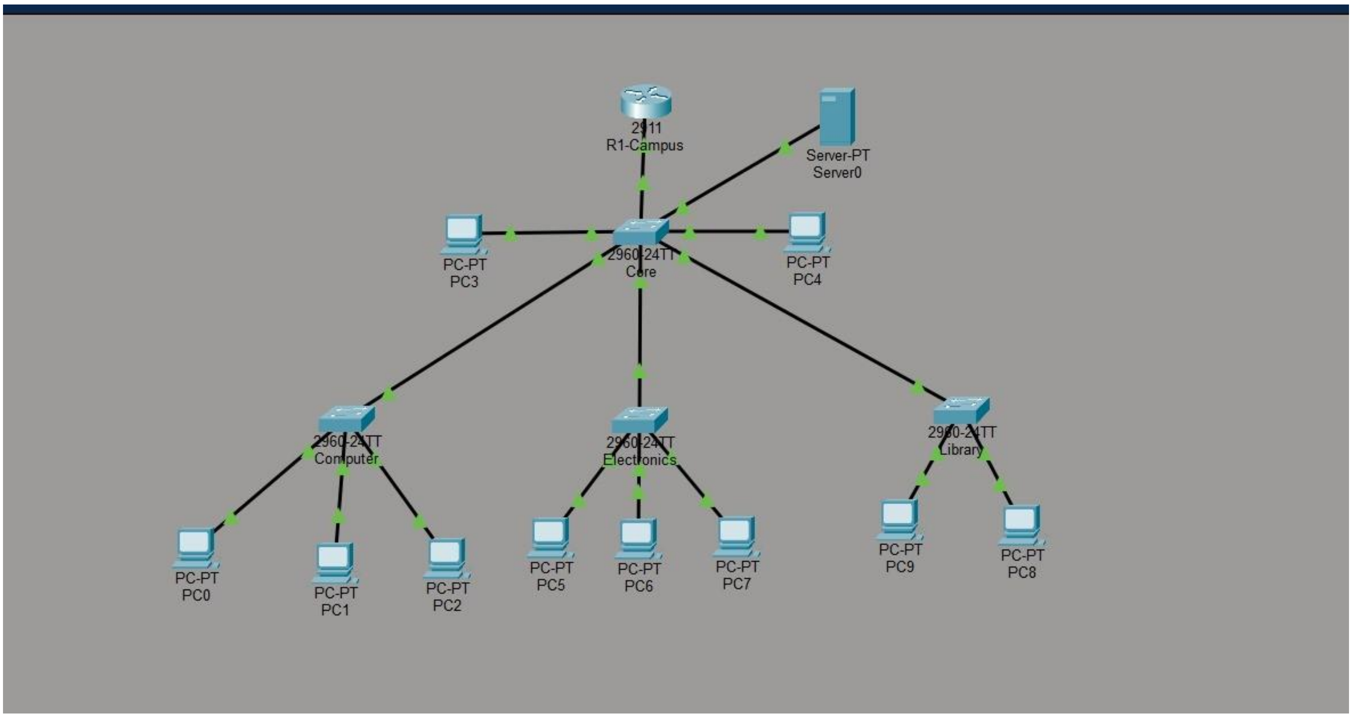

Figure 2: Router running configuration

The reference report records the router configuration with GigabitEthernet0/0 configured as 192.168.10.1/24 and GigabitEthernet0/

```
interface GigabitEthernet0/0
 ip address 192.168.10.1 255.255.255.0
 duplex
 auto
 speed
 auto

interface GigabitEthernet0/1
 ip address 192.168.20.1 255.255.255.0
 duplex
 auto
 speed
 auto
```

Additional evidence: PC0 connectivity tests, PC2-to-printer ping showing 4/4 replies, and the intentional wrong-gateway troubleshoo

10. Result

The campus network was successfully implemented in Cisco Packet Tracer. Both router interfaces were verified as up/up, devices

11. Conclusion

The campus network was successfully designed, simulated, configured, tested, and troubleshooted using Cisco Packet Tracer. The

Appendix – Contents from the Provided Reference File

The report retains the structure and contents of the supplied reference PDF, with the student group updated as requested.

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